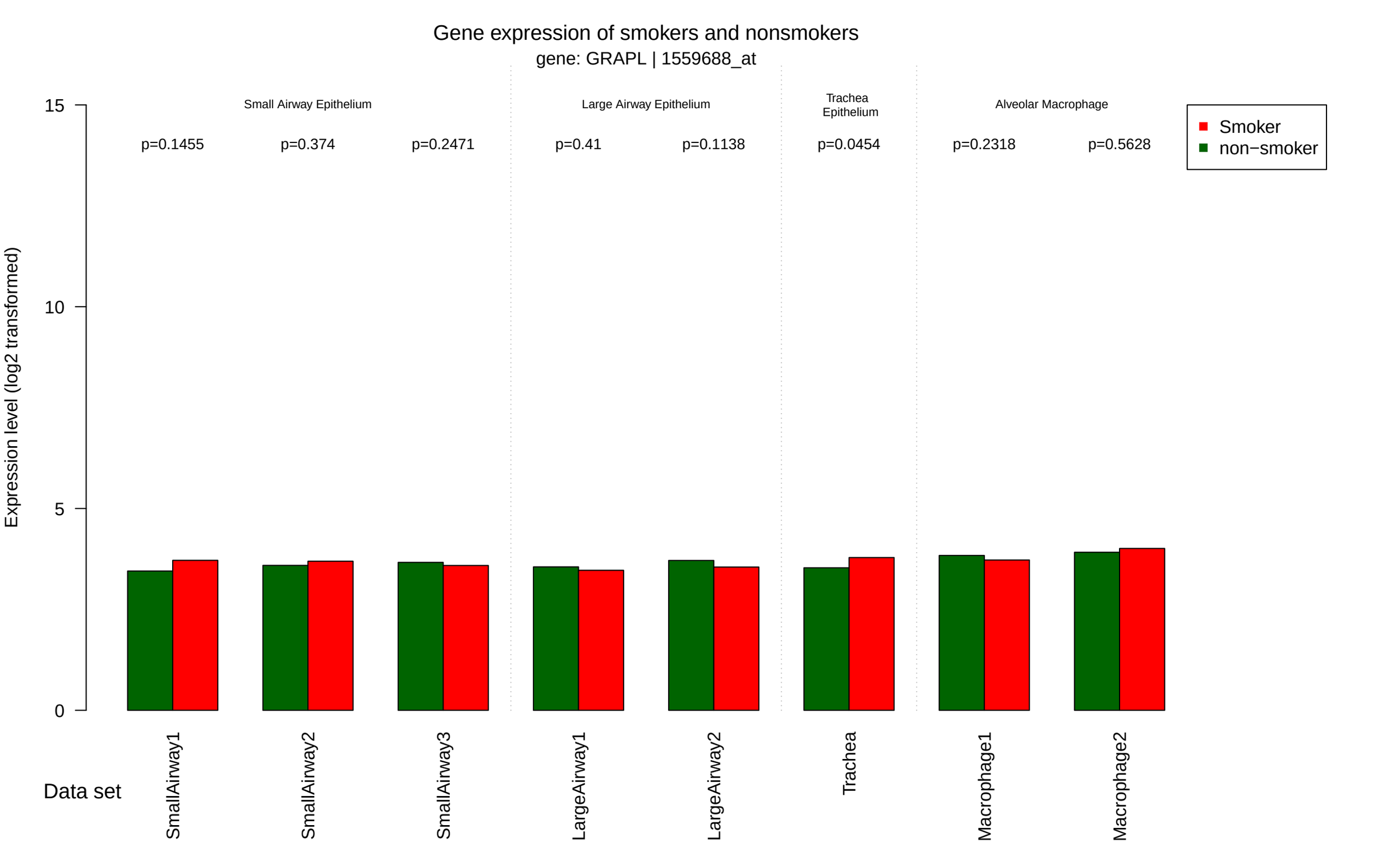
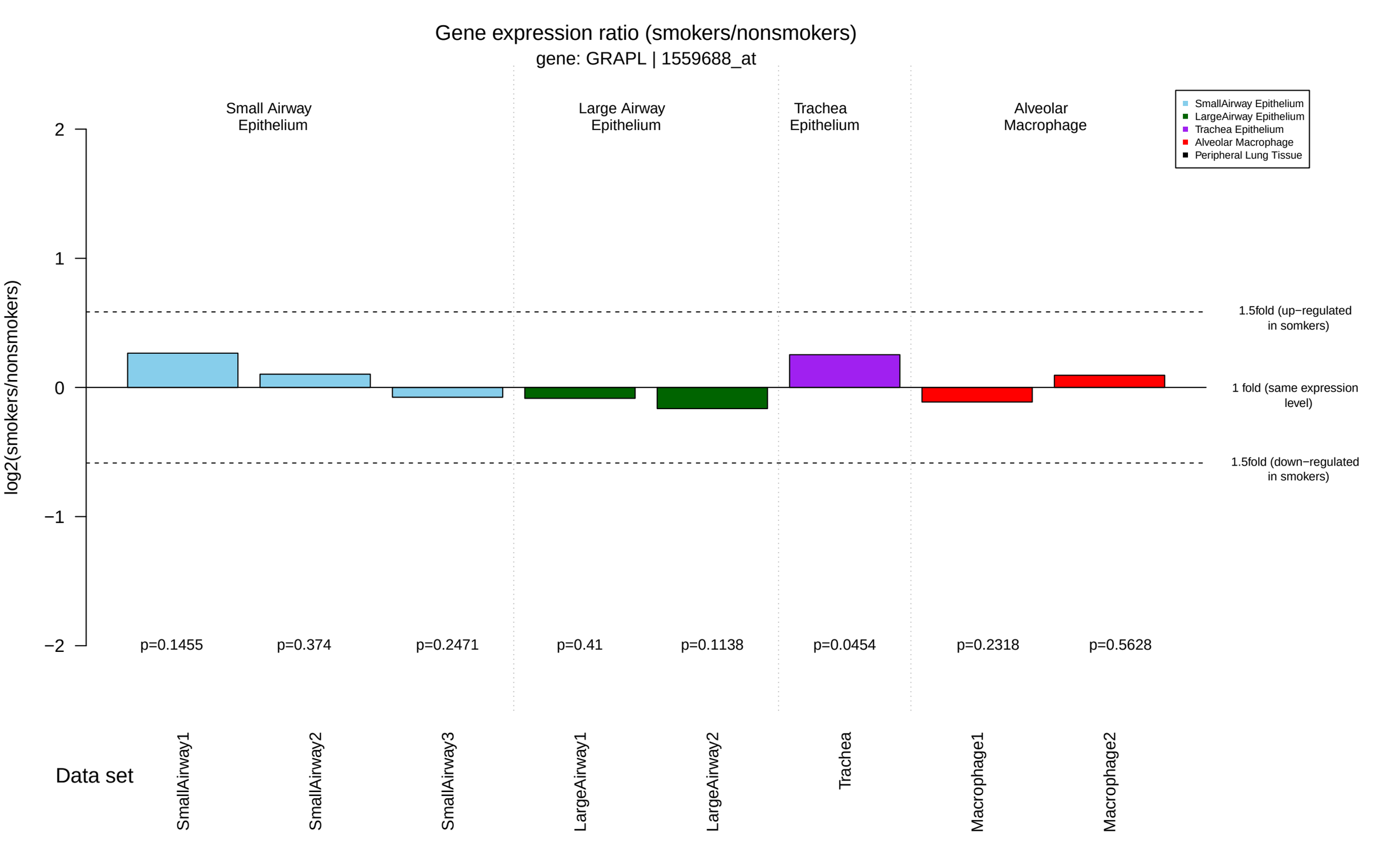
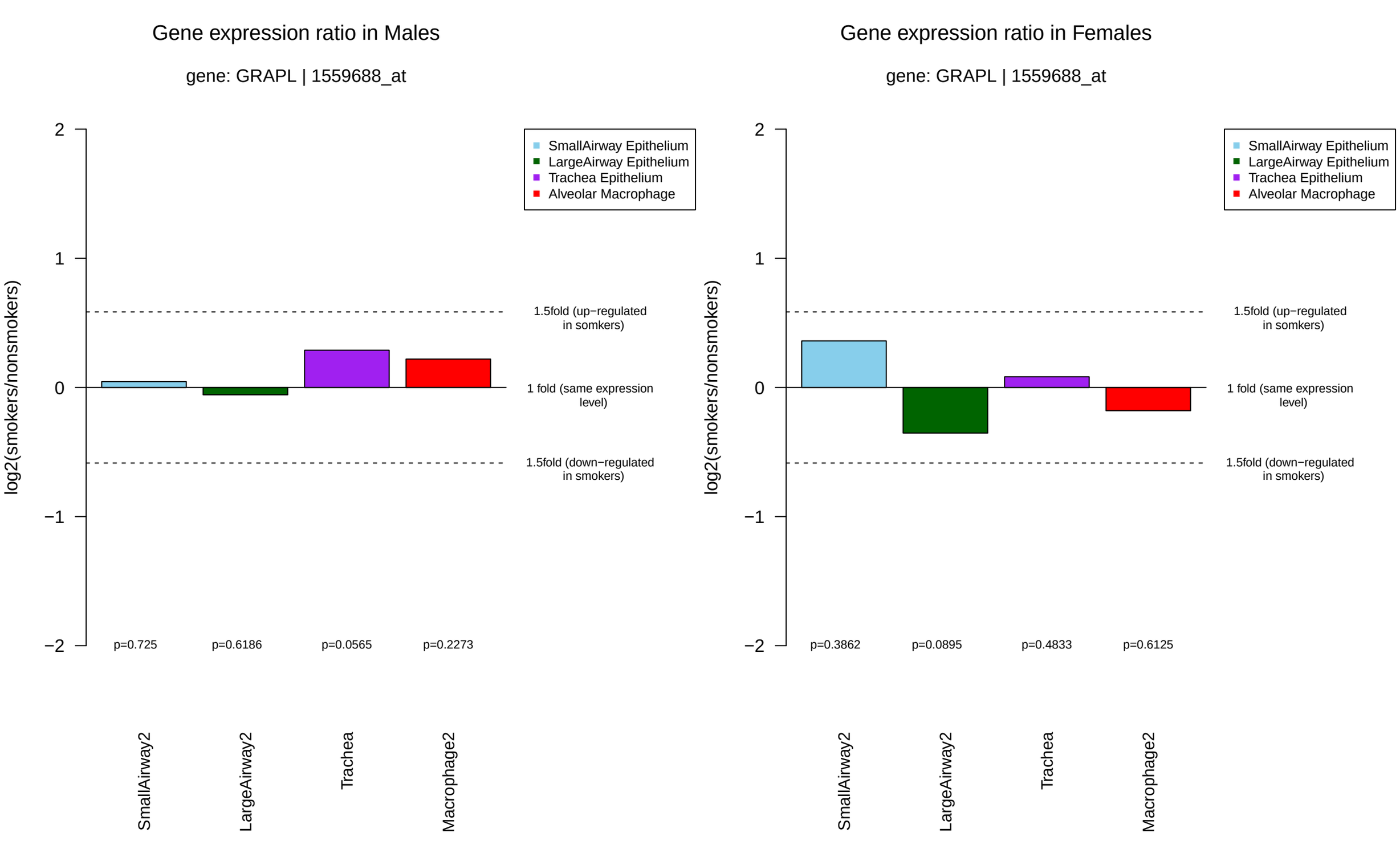
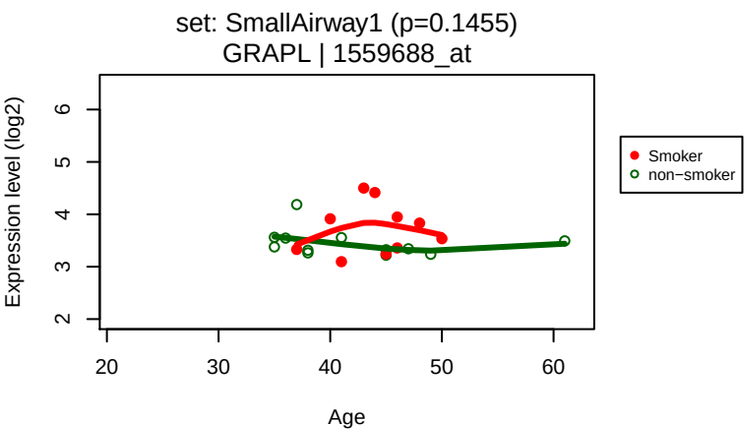
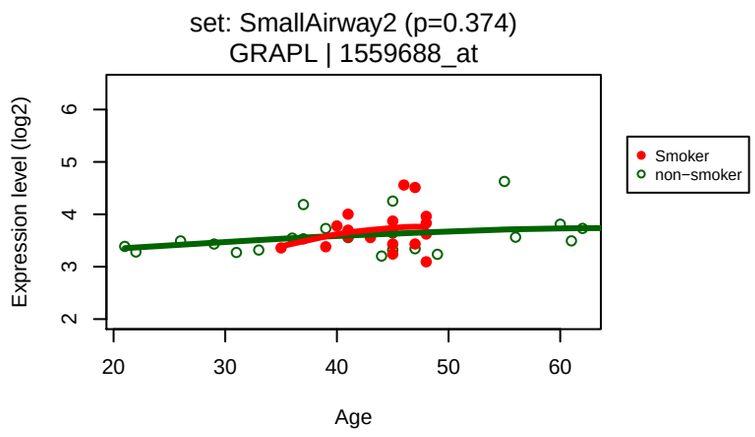
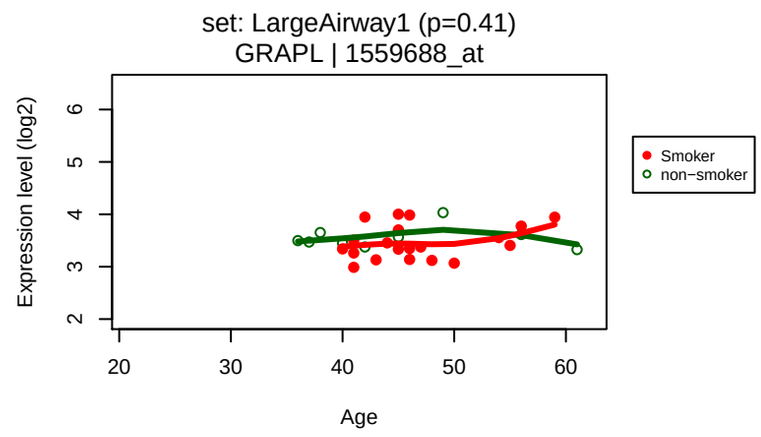

Gene expression of smokers and nonsmokers
gene: GRAPL | 1559688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1455
p=0.374
p=0.2471
p=0.41
p=0.1138
p=0.0454
p=0.2318
p=0.5628
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

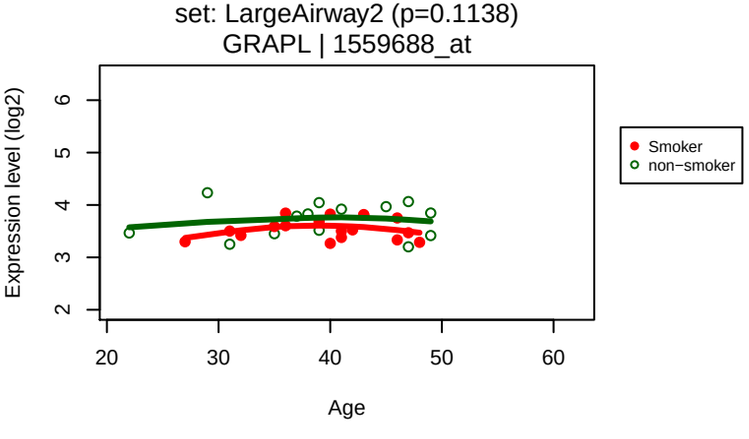

Gene expression ratio (smokers/nonsmokers)
gene: GRAPL | 1559688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1455
p=0.374
p=0.2471
p=0.41
p=0.1138
p=0.0454
p=0.2318
p=0.5628
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

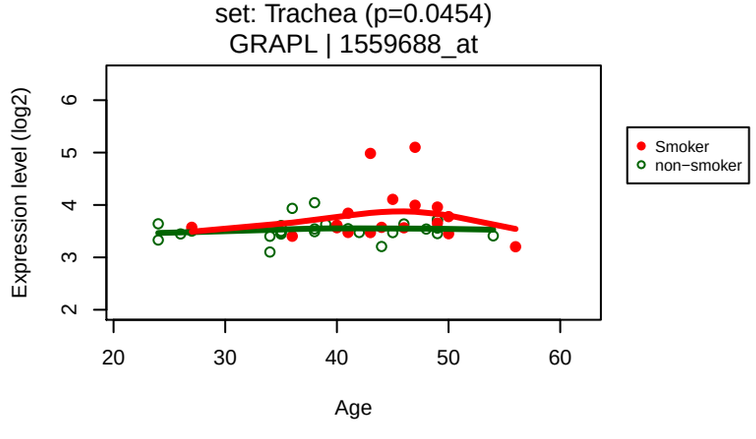

Gene expression ratio in Males
gene: GRAPL | 1559688_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.725
p=0.6186
p=0.0565
p=0.2273
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAPL | 1559688_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3862
p=0.0895
p=0.4833
p=0.6125
SmallAirway2
LargeAirway2
Trachea
Macrophage2

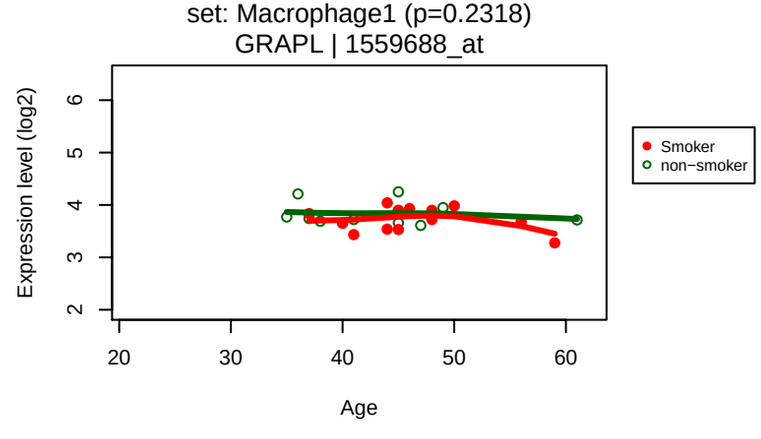

set: SmallAirway1 (p=0.1455)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker

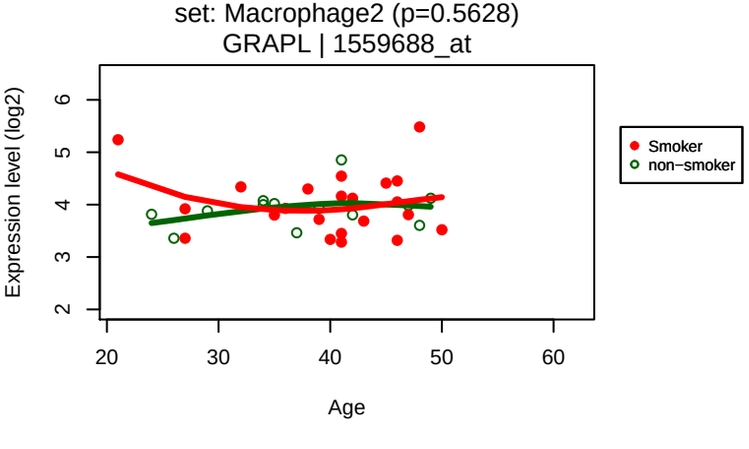

set: SmallAirway2 (p=0.374)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker

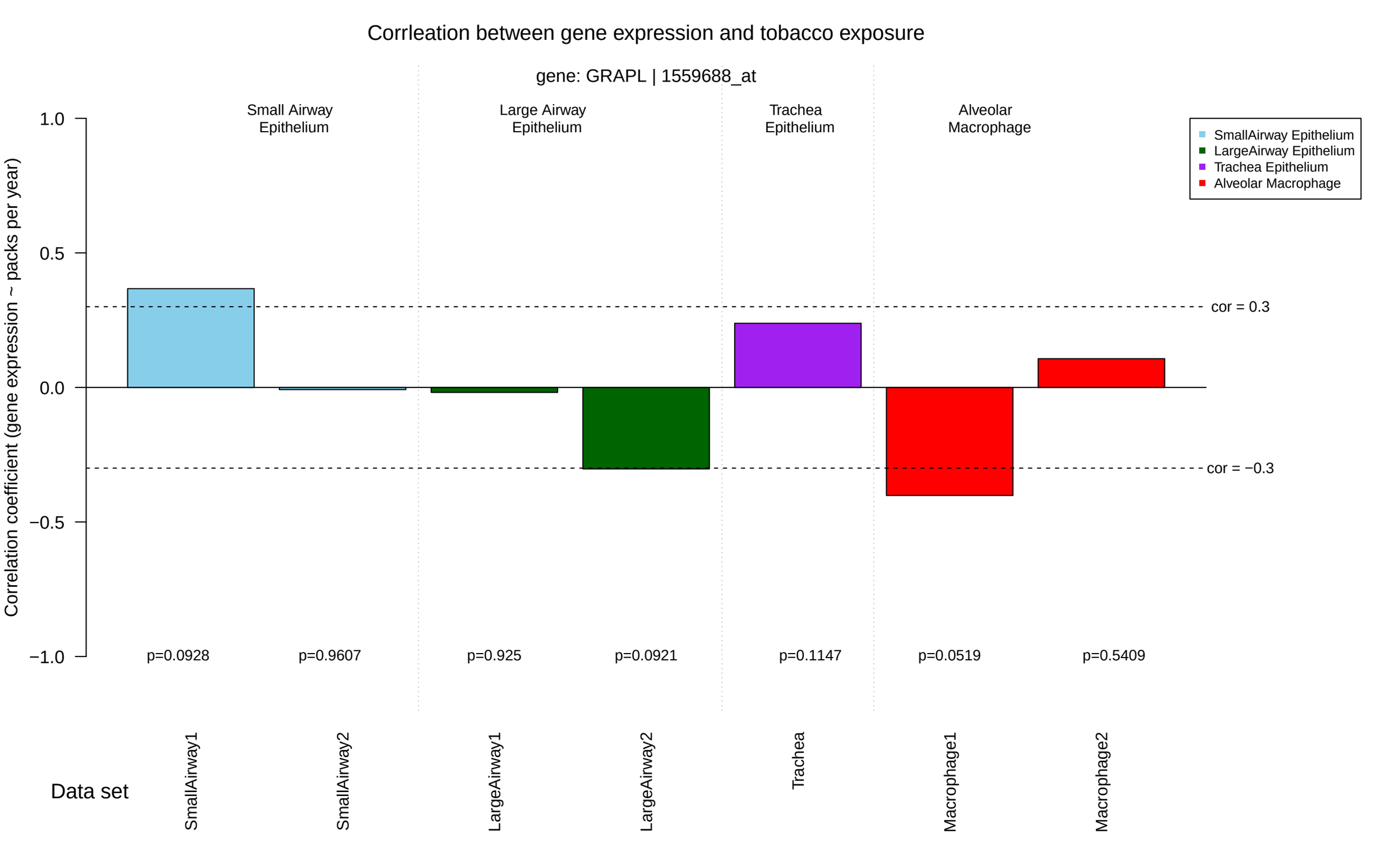

set: LargeAirway1 (p=0.41)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1138)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0454)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2318)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5628)
GRAPL | 1559688_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAPL | 1559688_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0928
p=0.9607
p=0.925
p=0.0921
p=0.1147
p=0.0519
p=0.5409
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2